

1.2 Measurements & Errors

Question Paper

Course	CIEA Level Physics
Section	1. Physical Quantities & Units
Topic	1.2 Measurements & Errors
Difficulty	Hard

Time allowed: 20

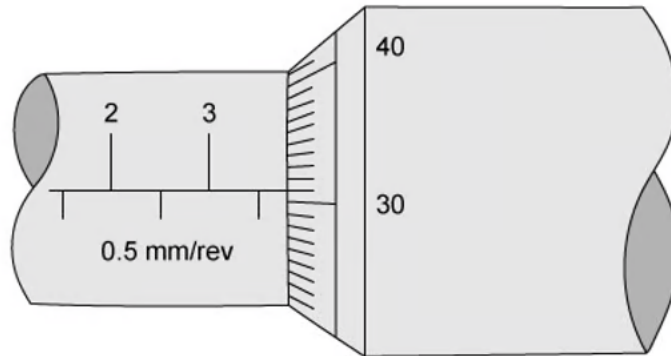
Score: /10

Percentage: /100

Question 1

The diameter of a cylindrical metal rod is measured using a micrometer screw gauge.

The diagram below shows an enlargement of the scale on the micrometer screw gauge when taking the measurement



What is the cross-sectional area of the rod?

- A. 3.81 mm^2
- B. 11.4 mm^2
- C. 22.8 mm^2
- D. 45.6 mm^2

[1 mark]

Question 2

A student uses a digital ammeter to measure a current. The reading of the ammeter is found to fluctuate between 1.98 A and 2.02 A.

The manufacturer of the ammeter states that any reading has a systematic uncertainty of $\pm 1\%$.

Which value of current should be quoted by the student?

- A. $(2.00 \pm 0.01) \text{ A}$
- B. $(2.00 \pm 0.02) \text{ A}$
- C. $(2.00 \pm 0.03) \text{ A}$
- D. $(2.00 \pm 0.04) \text{ A}$

[1 mark]

Question 3

The uncertainty in the value of the momentum of a trolley passing between two points X and Y varies with the choice of measuring devices.

Measurements for the same trolley made by different instruments were recorded.

- 1 distance between X and Y using a metre rule with cm divisions = 0.55 m
- 2 distance between X and Y using a metre rule with mm divisions = 0.547 m
- 3 timings using a wristwatch measuring to the nearest 0.5 s at X = 0.0 s and at Y = 4.5 s
- 4 timings using light gates measuring to the nearest 0.1 s at X = 0.0 s and at Y = 4.3 s
- 5 mass of trolley using a balance measuring to the nearest g = 6.4×10^{-2} kg
- 6 mass of trolley using a balance measuring to the nearest 10 g = 6×10^{-2} kg

Which measurements, one for each quantity measured, leads to the least uncertainty in the value of the momentum of the trolley?

- A. 1, 3 and 6
- B. 1, 4 and 6
- C. 2, 3 and 6
- D. 2, 4 and 5

[1 mark]

Question 4

In an experiment to determine the acceleration of free fall g , a ball bearing is held by an electromagnet. When the current to the electromagnet is switched off, a clock starts and the ball bearing falls. After falling a distance h , the ball bearing strikes a switch to stop the clock which measures the time t of the fall.

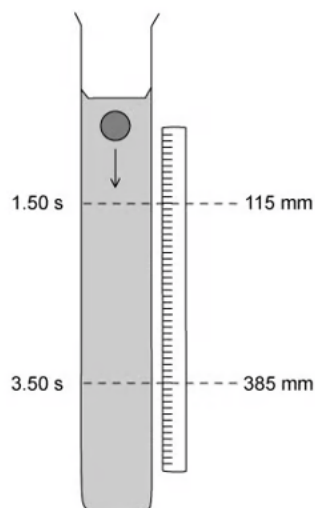
If systematic errors cause t and h to be measured incorrectly, which error **must** cause g to appear greater than 9.81 m s^{-2} ?

- A. h measured as being **smaller** than it actually is and t is measured correctly
- B. h measured as being **smaller** than it actually is and t measured as being **larger** than it actually is
- C. h measured as being **larger** than it actually is and t measured as being **larger** than it actually is
- D. h is measured correctly and t measured as being **smaller** than it actually is

[1 mark]

Question 5

The diagram shows an experiment to measure the speed of a small ball falling at constant speed through a clear liquid in a glass tube.



There are two marks on the tube. The top mark is positioned at 115 ± 1 mm on the adjacent rule and the lower mark at 385 ± 1 mm. The ball passes the top mark at 1.50 ± 0.02 s and passes the lower mark at 3.50 ± 0.02 s.

The constant speed of the ball is calculated to be 135 mm s^{-1}

Which expression calculates the fractional uncertainty in the value of this speed?

A. $\frac{2}{270} + \frac{0.04}{2.00}$

B. $\frac{1}{270} + \frac{0.02}{2.00}$

C. $\frac{1}{115} + \frac{0.02}{1.50}$

D. $\frac{1}{385} + \frac{0.02}{3.50}$

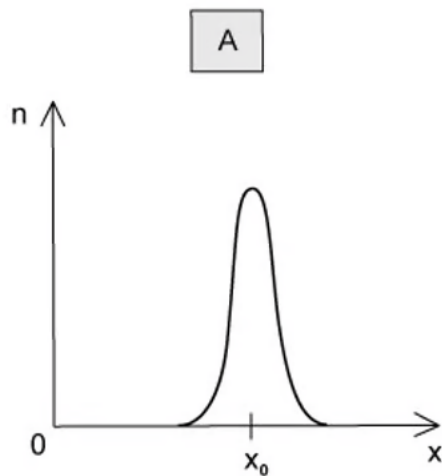
[1 mark]

Question 6

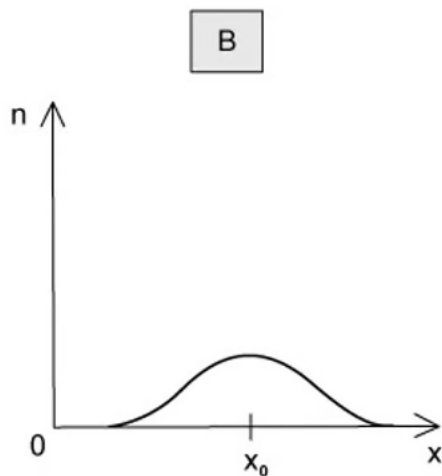
A fixed quantity x_0 is measured many times in an experiment that has experimental uncertainty. A graph is plotted to show the number n of times that a particular value x is obtained.

Which graph could be obtained if the measurement of x_0 has a large systematic error but a small random error?

A.

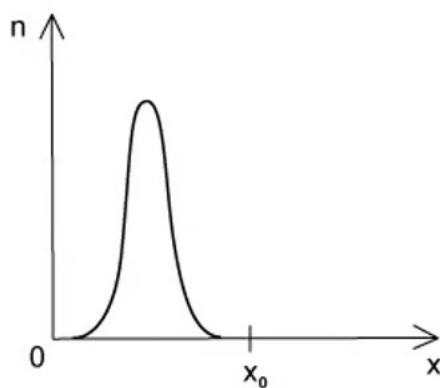


B.



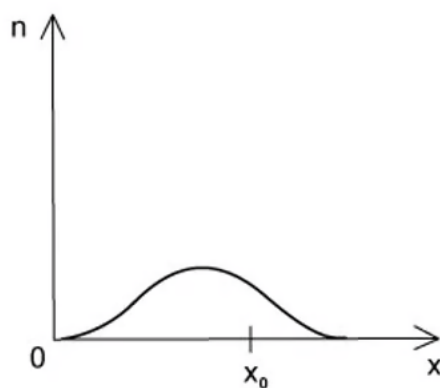
C.

C



D.

D



[1 mark]

Question 7

A micrometer screw gauge is used to measure the diameter of a copper wire.

The reading with the wire in position is shown in diagram 1. The wire is removed and the jaws of the micrometer are closed. The new reading is shown in diagram 2.

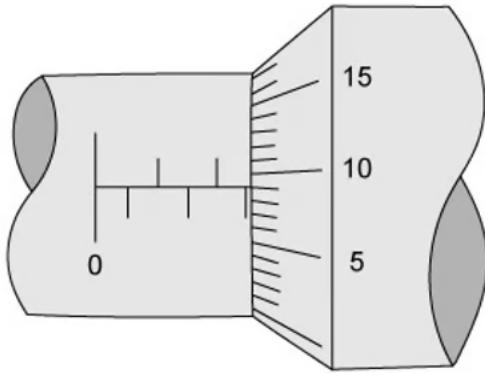


Diagram 1

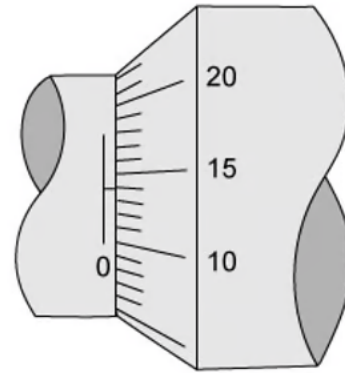


Diagram 2

What is the diameter of the wire?

- A. 1.90 mm
- B. 2.45 mm
- C. 2.59 mm
- D. 2.73 mm

[1 mark]

Question 8

A student finds the density of a liquid by measuring its mass and volume. The following is a summary of his measurements.

mass of empty beaker = (20 ± 1) g

mass of beaker + liquid = (70 ± 1) g

volume of liquid = (10.0 ± 0.5) cm³

He correctly calculates the density of the liquid as 5.0 g cm^{-3} .

What is the uncertainty in this value?

- A. 0.3 g cm^{-3}
- B. 0.5 g cm^{-3}
- C. 0.6 g cm^{-3}
- D. 2.6 g cm^{-3}

[1 mark]

Question 9

A steel wire is stretched in an experiment to determine the Young modulus for steel.

The uncertainties in the measurements are given below.

Measurement	Uncertainty
load on wire	$\pm 2\%$
length of wire	$\pm 0.2\%$
diameter of wire	$\pm 1.5\%$
extension	$\pm 1\%$

What is the percentage uncertainty in the Young modulus?

- A. 1.3%
- B. 1.8%
- C. 4.7%
- D. 6.2%

[1 mark]

Question 10

A mass is dropped from rest and falls through a distance of 2.0 m in a vacuum. An observer records the time taken for the mass to fall through this distance using a manually operated stopwatch and repeats the measurements a further two times. The average result of these measured times, displayed in the table below, was used to determine a value for the acceleration of free fall. This was calculated to be 9.8 m s^{-2} .

	First measurement	Second measurement	Third measurement	Average
Time/s	0.6	0.73	0.59	0.64

Which statement best relates to the experiment?

- A. The measurements are precise and accurate with no evidence of random errors.
- B. The measurements are not accurate and not always recorded to the degree of precision of the measuring device but the calculated experimental result is accurate.
- C. The measurements are not always recorded to the degree of precision of the measuring device but are accurate. Systematic errors may be present.
- D. The range of results shows that there were random errors made but the calculated value is correct so the experiment was successful.

[1 mark]

Model Answers: Hard

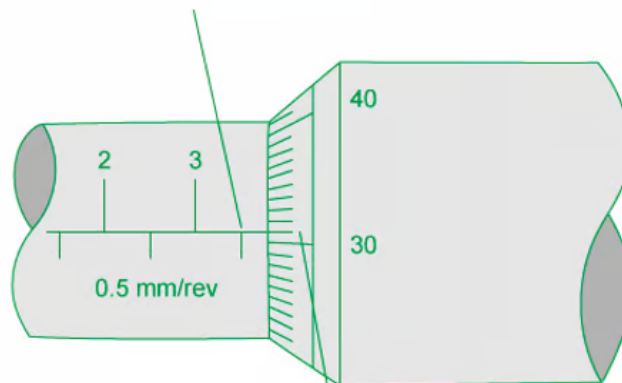
1

The correct answer is **B** because:

- The main scale reading is 3.5 mm
- The thimble scale reading is 0.31 mm
- So the diameter of the rod is $3.5 + 0.31 = 3.81$ mm
- The cross-sectional area is equal to

$$\circ \pi r^2 = \pi \left(\frac{d}{2} \right)^2 = \pi \left(\frac{3.81}{2} \right)^2 = 11.4 \text{ mm}^2$$

Read the main scale to the nearest 0.5 mm, in this case it's 3.5 mm



Then read the number of divisions on the thimble, which will be in 0.01 mm, in this case it is $31 \times 0.01 = 0.31$ mm

2

The correct answer is **D** because:

- The mean value of the current is $\frac{1.98 + 2.02}{2} = 2.00 \text{ A}$

- The fluctuation between 1.98 A and 2.02 A is caused by random errors, so the random uncertainty is ± 0.02 A
- The systematic uncertainty is $\pm 1\%$ which is $2.00 \times \frac{1}{100} = 0.02$ A
- The total uncertainty = random uncertainty + systematic uncertainty
 - $0.02 + 0.02 = \pm 0.04$ A
- So the current quoted should be 2.00 ± 0.04 A

3

The correct answer is **D** because:

- Looking at the equation for momentum
 - $p = mv = \frac{ms}{t}$
 - $\frac{\Delta p}{p} = \frac{\Delta m}{m} + \frac{\Delta s}{s} + \frac{\Delta t}{t}$
- We can see that for the uncertainty in momentum (Δp) to have the least value, the uncertainties in all the quantities should be as small as possible
- The instruments which measure to a higher degree of precision will lead to less uncertainty in the value of momentum
 - The metre rule in 2 has a greater precision than 1 because the smallest scale division is in mm which is more accurate than cm.
 - The light gate in 4 has a greater precision than the wristwatch in 3 because 4.3 s is more accurate than 4.5 s
 - The balance which measures to the nearest g has a greater precision than the balance which measures to the nearest 10 g as it measures to an extra decimal point making the measurement more accurate

4

The correct answer is **D** because:

- To determine the relationship between g , h and t we will need to consider the equation
 - $s = ut + \frac{1}{2}at^2$
- Since the falling body will initially be at rest, $u = 0$, the equation simplifies to

$$\circ h = \frac{1}{2} g t^2$$

- Rearranging the equation above we can see

$$\circ g = \frac{2h}{t^2}$$

- We can see that g and h are directly proportional so as h increases, g increases and vice versa
- We can see that g and t are inversely proportional, so as t increases, g decreases and vice versa
- Therefore, if h is measured correctly, g will be calculated correctly but if t is measured to be lower than it should this will cause g to be larger than it should be

A is incorrect because this would lead to a value of g which is smaller than it should be

B & C are incorrect because without knowing how much larger or smaller these quantities are, it's not possible to say if g will be larger or smaller

5

The correct answer is **A** because:

- The equation relating speed, distance and time is given by

$$\circ \text{speed } (v) = \frac{\text{distance } (s)}{\text{time } (t)}$$
- The value and uncertainty in distance is:
 - $\circ s = 385 - 115 = 270 \text{ mm}$
 - $\circ \Delta s = \pm(1 + 1) = \pm 2 \text{ mm}$
- The value and uncertainty in time is:
 - $\circ t = 3.50 - 1.50 = 2.00 \text{ s}$
 - $\circ \Delta t = \pm(0.02 + 0.02) = \pm 0.04 \text{ s}$
- To work out the fractional uncertainty in speed:
 - $\circ \frac{\Delta v}{v} = \frac{\Delta s}{s} + \frac{\Delta t}{t}$
 - $\circ \frac{\Delta v}{v} = \frac{2}{270} + \frac{0.04}{2.00}$

6

The correct answer is **C** because:

- Random errors affect precision, meaning they cause differences in measurements which causes a spread about the mean. Graph **C** shows a narrow curve, with a high peak so the measurements are close to each other, meaning the random error is low
- Systematic error affects accuracy, which causes all results to be too high or too low by the same amount each time. Graph **C** shows the measurements are consistent but they are much lower meaning the systematic error is high

A is incorrect because this graph shows a set of results with both low random and systematic errors

B is incorrect because this graph shows a set of results which have a low systematic error and a high random error

D is incorrect because this graph shows a set of results with both high random and systematic errors

7

The correct answer is **B** because:

- The main scale reading is 2.5 mm
- The thimble scale reading is 0.09 mm
- So the diameter of the wire is $2.5 + 0.09 = 2.59$ mm
- When the jaws are closed, it does not read zero instead there is a reading of 0.14 mm, so any measurement made must subtract this value to get the true diameter, which is
 - $2.59 - 0.14 = 2.45$ mm

$$\begin{aligned} \circ \quad \frac{\Delta v}{v} &= \frac{\Delta s}{s} + \frac{\Delta t}{t} \\ \circ \quad \frac{\Delta v}{v} &= \frac{2}{270} + \frac{0.04}{2.00} \end{aligned}$$

6

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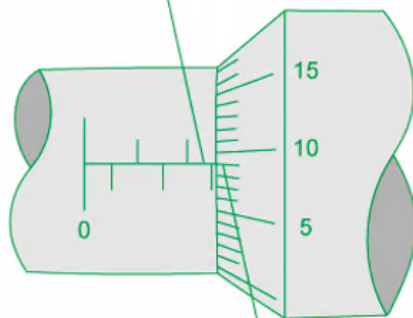
D is incorrect because this graph shows a set of results with both high random and systematic errors

7

The correct answer is **B** because:

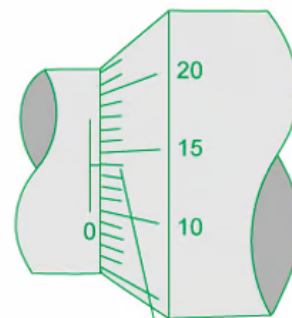
- The main scale reading is 2.5 mm
- The thimble scale reading is 0.09 mm
- So the diameter of the wire is $2.5 + 0.09 = 2.59$ mm
- When the jaws are closed, it does not read zero instead there is a reading of 0.14 mm, so any measurement made must subtract this value to get the true diameter, which is
 - $2.59 - 0.14 = 2.45$ mm

Read the main scale to the nearest 0.5 mm, in this scale it is 2.5 mm



Then read the number of divisions on the thimble scale, which will be in 0.1 mm in this case it is $9 \times 0.01 = 0.09$ mm

Diagram 1



When the jaws are closed there is a reading of $14 \times 0.01 = 0.14$ mm

Diagram 2

8

The correct answer is **B** because:

- The equation relating density, mass and volume is given by

$$\text{density } (\rho) = \frac{\text{mass}(m)}{\text{volume}(v)}$$

- The value and uncertainty in mass is:

$$m = 70 - 20 = 50 \text{ g}$$

$$\Delta m = \pm(1 + 1) = \pm 2 \text{ g}$$

- The value and uncertainty in volume is:

$$V = 10 \text{ cm}^3$$

$$\Delta V = \pm 0.5 \text{ cm}^3$$

- To work out the uncertainty in density:

$$\frac{\Delta \rho}{\rho} = \frac{\Delta m}{m} + \frac{\Delta V}{V}$$

$$\frac{\Delta \rho}{\rho} = \frac{2}{50} + \frac{0.5}{10} = 0.04 + 0.05 = 0.09$$

- $\rho = 50 \pm 10 \text{ g cm}^{-3}$
 ○ $\frac{\Delta\rho}{\rho} = 0.09$, where $\rho = 5.0 \text{ g cm}^{-3}$ so
 ○ $\Delta\rho = 5 \times 0.09 = 0.45 = 0.5 \text{ g cm}^{-3}$ (2 s.f)

9

The correct answer is **D** because:

- The equation for Young Modulus is:

$$\text{Young Modulus} = \frac{\text{Force}(F)}{\text{Area}(A)} \times \frac{\text{length}(l)}{\text{extension}(x)} = \frac{\text{load} \times \text{length}}{\frac{\pi}{4} \times \text{diameter}^2 \times \text{extension}}$$
- To find the percentage uncertainty in Young Modulus we must add the uncertainties in the measurements
- We can ignore the $\frac{\pi}{4}$ since it is a constant with no units, and it is important to multiply the uncertainty in the diameter by 2 since the diameter is squared
 - % uncertainty = $2 + 0.2 + (2 \times 1.5) + 1 = 6.2\%$

10

The correct answer is **B** because:

- Precision is how close the measured values are to each other
- In a measuring instrument, the precision is indicated by the number of significant figures given by the instrument
- From the measurements, we can see that the readings are not very close to each other. This scatter of readings indicates the presence of random errors
- Even though the same instrument is used, the measurements are not given to the same number of decimal places, so the measurements are not precise
- Accuracy is how close a measured value is to the true value
- We can see that the individual measurements are not close to the average value of time. So, the measurements are not accurate, but the calculated experimental is correct

A is incorrect because even though the correct experimental value was achieved, the results do not show accuracy or precision so the experiment was not successful

C is incorrect because systematic error affects accuracy and occurs due to the apparatus used or faults in the experimental method, so if systematic errors are present then the measurements are not accurate

D is incorrect because the significant random errors indicate that the experiment was not successful, even though the calculated value is close to the actual value

